Exercise 1

For the 3R spatial robot in Fig. 1, determine the symbolic expression of the elements of the inertia matrix $\boldsymbol{B}(\boldsymbol{q})$ using the recursive algorithm with moving frames for the computation of the kinetic energy of the links. The coordinates $\boldsymbol{q} \in \mathbb{R}^3$ to be used are those of the Denavit-Hartenberg (DH) convention.

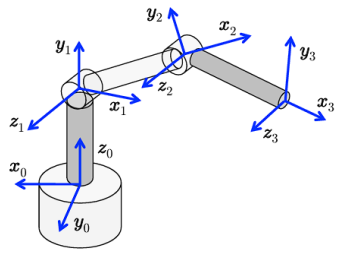
Besides the masses m_i, $i = 1, 2, 3$, of the links, the other constant dynamic parameters are specified as

$$ {}^1\boldsymbol{r}_{1,c1} = \begin{pmatrix} A \\ -F \\ 0 \end{pmatrix}, \qquad {}^2\boldsymbol{r}_{2,c2} = \begin{pmatrix} -C \\ 0 \\ 0 \end{pmatrix}, \qquad {}^3\boldsymbol{r}_{3,c3} = \begin{pmatrix} -D \\ 0 \\ E \end{pmatrix}, \tag{1} $$

where A, C, D, E and F take positive values, and

$$ {}^i\boldsymbol{I}_{ci} = \begin{pmatrix} I_{xx,i} & 0 & 0 \\ 0 & I_{yy,i} & 0 \\ 0 & 0 & I_{zz,i} \end{pmatrix}, \qquad i = 1, 2, 3. \tag{2} $$

Once $\boldsymbol{B}(\boldsymbol{q})$ has been obtained, define a set of dynamic coefficients that linearly parametrize the inertia matrix and is of the smallest possible cardinality.



i	α_i	d_i	a_i	θ_i
1	$\pi/2$	L_1	0	q_1
2	0	0	L_2	q_2
3	0	0	L_3	q_3

Firstly, i compute the homogeneus transformations.

$$ {}^0T_1 = \begin{pmatrix} c_1 & 0 & s_1 & 0 \\ s_1 & 0 & -c_1 & 0 \\ 0 & 1 & 0 & L_1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad {}^1T_2 = \begin{pmatrix} c_2 & -s_2 & 0 & L_2 c_2 \\ s_2 & c_2 & 0 & L_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad {}^2T_3 = \begin{pmatrix} c_3 & -s_3 & 0 & L_3 c_3 \\ s_3 & c_3 & 0 & L_3 s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} $$

then, i need to compute ${}^i v_{ci}, {}^i\omega_i$ for $i = 1, 2, 3$ by using:

$$ {}^i\omega_i = {}^{i-1}R_i^T \left[{}^{i-1}\omega_{i-1} + \dot{q}_i (0 \;\; 0 \;\; 1)^T \right] \quad \text{and} \quad {}^i v_i = {}^{i-1}R_i^T \left[{}^{i-1}v_{i-1} + {}^{i-1}\omega_{i-1} \times {}^{i-1}r_{i-1,i} \right] $$

and

$$ {}^i v_{c_i} = {}^i v_i + \underbrace{{}^i\omega_i \times r_{c_i}}_{\begin{pmatrix} 0 & -\omega_{iz} & \omega_{iy} \\ \omega_{iz} & 0 & -\omega_{ix} \\ -\omega_{iy} & \omega_{ix} & 0 \end{pmatrix} r_{c_i}} $$

$i=1$)

$$ {}^1\omega_1 = \begin{pmatrix} c_1 & s_1 & 0 \\ 0 & 0 & 1 \\ s_1 & -c_1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \dot{q}_1 \end{pmatrix} = \begin{pmatrix} 0 \\ \dot{q}_1 \\ 0 \end{pmatrix} \qquad {}^1 v_1 = \begin{pmatrix} c_1 & s_1 & 0 \\ 0 & 0 & 1 \\ s_1 & -c_1 & 0 \end{pmatrix} \left[\begin{pmatrix} 0 \\ 0 \\ \dot{q}_1 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ L_1 \end{pmatrix} \right] = 0 $$

$$ {}^1 v_{c1} = \begin{pmatrix} 0 \\ \dot{q}_1 \\ 0 \end{pmatrix} \times \begin{pmatrix} A \\ -F \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & \dot{q}_1 \\ 0 & 0 & 0 \\ -\dot{q}_1 & 0 & 0 \end{pmatrix} \begin{pmatrix} A \\ -F \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ -A\dot{q}_1 \end{pmatrix} $$

$$ \Rightarrow T = \frac{1}{2} m_1 v_{c_1}^T v_{c_1} + \frac{1}{2} {}^1\omega_1^T I_{c_1} {}^1\omega_1 = \frac{1}{2} m_1 A^2 \dot{q}_1^2 + \frac{1}{2} I_{1,xx} \dot{q}_1^2 = \frac{1}{2} \dot{q}_1^2 \left(m_1 A^2 + I_{1xx} \right) $$

$i=2$)

$$ {}^2\omega_2 = \begin{pmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \left[\begin{pmatrix} 0 \\ \dot{q}_1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \dot{q}_2 \end{pmatrix} \right] = \begin{pmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ \dot{q}_1 \\ \dot{q}_2 \end{pmatrix} = \begin{pmatrix} \dot{q}_1 s_2 \\ \dot{q}_1 c_2 \\ \dot{q}_2 \end{pmatrix} $$

$$ {}^2 v_2 = \begin{pmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \left[\begin{pmatrix} 0 \\ \dot{q}_1 \\ \dot{q}_2 \end{pmatrix} \times \begin{pmatrix} L_2 c_2 \\ L_2 s_2 \\ 0 \end{pmatrix} \right] = \begin{pmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -L_2 s_2 \dot{q}_2 \\ L_2 c_2 \dot{q}_2 \\ -L_2 s_2 \dot{q}_1 \end{pmatrix} = \begin{pmatrix} 0 \\ L_2 \dot{q}_2 \\ -L_2 s_2 \dot{q}_1 \end{pmatrix} $$

$$ v_{c_2} = \begin{pmatrix} 0 \\ L_2 \dot{q}_2 \\ -L_2 s_2 \dot{q}_1 \end{pmatrix} + \begin{pmatrix} 0 & -\dot{q}_2 & \dot{q}_1 c_2 \\ \dot{q}_2 & 0 & -\dot{q}_1 s_2 \\ -\dot{q}_1 c_2 & \dot{q}_1 s_2 & 0 \end{pmatrix} \begin{pmatrix} -c \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \dot{q}_2 (L_2 - c) \\ \dot{q}_1 (-c c_2 - L_2 s_2) \end{pmatrix} $$

$$ T_2 = \frac{1}{2} \left[m_2 \dot{q}_2^2 (L_2 - c)^2 + m_2 \dot{q}_1^2 (-c c_2 - L_2 s_2)^2 + I_{2xx} \dot{q}_1^2 s_2^2 + I_{2zz} \dot{q}_1^2 c_2^2 + I_{2zz} \dot{q}_2^2 \right] $$

$$ = \frac{1}{2} \dot{q}_1^2 \left(m_2 (-c c_2 - L_2 s_2)^2 + \left(I_{2xx} s_2^2 + I_{2zz} c_2^2 \right) \right) + \frac{1}{2} \dot{q}_2^2 \left(I_{2zz} + m_2 (L_2 - c)^2 \right) $$

$i=3)$

$$ {}^3\omega_3 = \begin{pmatrix} c_3 & s_3 & 0 \\ -s_3 & c_3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \left[\begin{pmatrix} \dot q_1 s_2 \\ \dot q_1 c_2 \\ \dot q_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \dot q_3 \end{pmatrix} \right] = \begin{pmatrix} c_3 & s_3 & 0 \\ -s_3 & c_3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \dot q_1 s_2 \\ \dot q_1 c_2 \\ \dot q_2 + \dot q_3 \end{pmatrix} = \begin{pmatrix} \dot q_1 (s_2 c_3 + c_2 s_3) \\ \dot q_1 (c_2 c_3 - s_2 s_3) \\ \dot q_2 + \dot q_3 \end{pmatrix} = \begin{pmatrix} \dot q_1 s_{23} \\ \dot q_1 c_{23} \\ \dot q_2 + \dot q_3 \end{pmatrix} $$

$$ {}^3v_3 = \begin{pmatrix} c_3 & s_3 & 0 \\ -s_3 & c_3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \left[\begin{pmatrix} 0 \\ L_2 \dot q_2 \\ -L_2 s_2 \dot q_1 \end{pmatrix} + \begin{pmatrix} \dot q_1 s_2 \\ \dot q_1 c_2 \\ \dot q_2 + \dot q_3 \end{pmatrix} \times \begin{pmatrix} L_3 c_3 \\ L_3 s_3 \\ 0 \end{pmatrix} \right] $$

$$ = \begin{pmatrix} c_3 & s_3 & 0 \\ -s_3 & c_3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \left[\begin{pmatrix} 0 \\ L_2 \dot q_2 \\ -L_2 s_2 \dot q_1 \end{pmatrix} + \begin{pmatrix} -L_3 s_3 (\dot q_2 + \dot q_3) \\ L_3 c_3 (\dot q_2 + \dot q_3) \\ -\dot q_1 L_3 c_{23} \end{pmatrix} \right] = \begin{bmatrix} L_2 s_3 \dot q_2 \\ L_3 (\dot q_2 + \dot q_3) + L_2 c_3 \dot q_2 \\ -\dot q_1 L_3 c_{23} - L_2 s_2 \dot q_1 \end{bmatrix} $$

$$ v_{c3} = \begin{bmatrix} L_2 s_3 \dot q_2 \\ L_3 (\dot q_2 + \dot q_3) + L_2 c_3 \dot q_2 \\ -\dot q_1 L_3 c_{23} - L_2 s_2 \dot q_1 \end{bmatrix} + \begin{pmatrix} \dot q_1 s_{23} \\ \dot q_1 c_{23} \\ \dot q_2 + \dot q_3 \end{pmatrix} \times \begin{pmatrix} -D \\ 0 \\ E \end{pmatrix} = \begin{bmatrix} L_2 s_3 \dot q_2 + E \dot q_1 c_{23} \\ L_3 (\dot q_2 + \dot q_3) + L_2 c_3 \dot q_2 - D(\dot q_2 + \dot q_3) - E \dot q_1 s_{23} \\ D \dot q_1 c_{23} - \dot q_1 L_3 c_{23} - L_2 s_2 \dot q_1 \end{bmatrix} $$

I compute separately the terms of T_3

$$ \frac{1}{2} {}^3\omega_3^T I_{c3} {}^3\omega_3 = \frac{1}{2} \begin{pmatrix} \dot q_1 s_{23} & \dot q_1 c_{23} & \dot q_2 + \dot q_3 \end{pmatrix} \begin{pmatrix} I_{3xx} \dot q_1 s_{23} \\ I_{3yy} \dot q_1 c_{23} \\ I_{3zz} \dot q_2 + \dot q_3 \end{pmatrix} = \frac{1}{2} \left[I_{3xx} s_{23}^2 \dot q_1^2 + I_{3yy} c_{23}^2 \dot q_1^2 + I_{3zz} (\dot q_2 + \dot q_3)^2 \right] $$

$$ \begin{aligned} v_{c3}^T v_{c3} = {} & L_2^2 s_3^2 \dot q_2^2 + E^2 \dot q_1^2 c_{23}^2 + 2 L_2 E s_3 c_{23} \dot q_1 \dot q_2 \\ & + (\dot q_2 + \dot q_3)^2 (L_3 - D)^2 + \dot q_2^2 L_2^2 c_3^2 - \dot q_1^2 s_{23}^2 + 2 (\dot q_2 + \dot q_3)(L_3 - D)(\dot q_2 L_2 c_3 - \dot q_1 s_{23}) \\ & + \dot q_1^2 \left[c_{23}^2 [D - L_3]^2 + L_2^2 s_2^2 - 2 L_2 [D - L_3] s_2 c_{23} \right. \end{aligned} $$

The total kinetick energy is

$$ \begin{aligned} 2 \cdot T = {} & \dot q_1^2 (m_1 A^2 + I_{1xx}) + \\ & \dot q_1^2 \left(m_2 (-c c_2 - L_2 s_2)^2 + (I_{2xx} s_2^2 + I_{2yy} c_2^2) \right) + \dot q_2^2 \left(I_{2zz} + m_2 (L_2 - c)^2 \right) \\ & + \left[I_{3xx} s_{23}^2 \dot q_1^2 + I_{3yy} c_{23}^2 \dot q_1^2 + I_{3zz} (\dot q_2 + \dot q_3)^2 \right] \\ & m_3 \left[L_2^2 s_3^2 \dot q_2^2 + E^2 \dot q_1^2 c_{23}^2 + 2 L_2 E s_3 c_{23} \dot q_1 \dot q_2 \right. \\ & + (\dot q_2 + \dot q_3)^2 (L_3 - D)^2 + \dot q_2^2 L_2^2 c_3^2 - \dot q_1^2 s_{23}^2 + 2 (\dot q_2 + \dot q_3)(L_3 - D)(\dot q_2 L_2 c_3 - \dot q_1 s_{23}) \\ & + \dot q_1^2 \left[c_{23}^2 [D - L_3]^2 + L_2^2 s_2^2 - 2 L_2 [D - L_3] s_2 c_{23} \right] \Big] \end{aligned} $$

$$2T = \dot{q}_1^2 \Big(m_2(-c c_2 - L_2 s_2)^2 + (I_{2xx} s_2^2 + I_{2zz} c_2^2) + m_1 A^2 + I_{1xx} + I_{3xx} s_{23}^2 + I_{3zz} c_{23}^2 + m_3 E^2 c_{23}^2$$

$$+ m_3 \left[c_{23}^2 [D - L_3]^2 + L_2^2 s_2^2 - 2 L_2 [D - L_3] s_2 c_{23} - s_{23}^2 \right] \Big) = M_{11}$$

$$+ \dot{q}_2^2 \left(I_{2zz} + m_2 (L_2 - c)^2 + I_{3zz} + m_3 + m_3 L_2^2 s_3^2 + m_3 L_2 c_3 \left(2(L_3 - D) + L_2 c_3 \right) \right) = M_{22}$$

$$+ \dot{q}_3^2 (I_{3zz} + m_3) = M_{33}$$

$$+ \dot{q}_2 \dot{q}_3 \left(I_{3zz} + m_3 2 L_2 c_3 (L_3 - D) + m_3 (L_3 - D)^2 \right) = 2M_{23} = 2M_{32}$$

$$+ \dot{q}_1 \dot{q}_2 \left(m_3 2 L_2 E s_3 c_{23} - 2 m_3 s_{23} (L_3 - D) \right) = 2M_{12} = 2M_{21}$$

$$\dot{q}_1 \dot{q}_3 \left(-2 m_3 s_{23} (L_3 - D) \right) = 2M_{13} = 2M_{31}$$

Exercise 2

When the Jacobian $\boldsymbol{J}(\boldsymbol{q})$ is a $m \times n$ matrix with full row rank m, its weighted pseudoinverse $\boldsymbol{J}_W^{\#}(\boldsymbol{q})$ takes the explicit form

$$\boldsymbol{J}_W^{\#}(\boldsymbol{q}) = \boldsymbol{W}^{-1} \boldsymbol{J}^T(\boldsymbol{q}) \left(\boldsymbol{J}(\boldsymbol{q}) \boldsymbol{W}^{-1} \boldsymbol{J}^T(\boldsymbol{q}) \right)^{-1}, \tag{3}$$

where $\boldsymbol{W}$ is a $n \times n$, symmetric, and positive definite matrix. The matrix $\boldsymbol{J}_W^{\#}$ in (3) satisfies three of the four identities that uniquely define a pseudoinverse. Prove that the weighted pseudoinverse takes the following more general form, which holds true even when the Jacobian $\boldsymbol{J}(\boldsymbol{q})$ loses rank:

$$\boldsymbol{J}_W^{\#}(\boldsymbol{q}) = \boldsymbol{W}^{-1/2} \left(\boldsymbol{J}(\boldsymbol{q}) \boldsymbol{W}^{-1/2} \right)^{\#}. \tag{4}$$

We have to prove the three identities. Firstly, i denote $J W^{-1/2} := A$

3) $$(J J_W^{\#})^T = J_W^{\#T} J^T = A^{\#T} \underbrace{W^{-1/2T} J^T}_{(JW^{-1/2})^T} = (JW^{-1/2})^{\#T} (JW^{-1/2})^T = (\underbrace{JW^{-1/2}}_{A} \cdot \underbrace{(JW^{-1/2})^{\#}}_{A^{\#}})^T = (AA^{\#})^T = AA^{\#}$$

$$= JW^{-1/2} \cdot (JW^{-1/2})^{\#} = J \left(W^{-1/2} (JW^{-1/2})^{\#} \right) = J J_W^{\#}. \quad \blacksquare$$

1) $$J J_W^{\#} J = \underbrace{J W^{-1/2} (JW^{-1/2})^{\#}}_{JJ_W^{\#} = (JJ_W^{\#})^T} J = \left((JW^{-1/2})^{\#} W^{-1/2} \right)^T J^T J = W^{-1/2} \left[(JW^{-1/2})^{\#} \right]^T J^T J = W^{-1/2} \left[J (JW^{-1/2})^{\#} \right]^T J$$

Exercise 3

Consider a 4R planar robot with all links of equal length $\ell = 0.5$ [m]. The robot is stretched along the $\boldsymbol{x}_0$ axis, in the DH configuration $\boldsymbol{q} = \boldsymbol{0}$. The end-effector of the robot should execute an instantaneous linear velocity $\boldsymbol{v} = \begin{pmatrix} 0 & 10 \end{pmatrix}^T$ [m/s]. The joint velocities are bounded as $|\dot{q}_i| \leq V_i$, $i = 1, \ldots, 4$, with $V_1 = 4$, $V_2 = 2$, and $V_3 = V_4 = 1$ [rad/s]. Find a feasible joint velocity $\dot{\boldsymbol{q}} \in \mathbb{R}^4$ that executes the given Cartesian task, either as such or in a scaled way, while satisfying the hard bounds on joint velocity. Scale the task velocity $\boldsymbol{v}$ only if strictly needed. A solution with a lower norm is preferred.

Hint: The following useful expressions hold for the pseudoinverse of a block matrix $\boldsymbol{A}$ (with a submatrix $\boldsymbol{O}$ of zeros) and of a vector $\boldsymbol{u} \neq \boldsymbol{0}$:

$$\boldsymbol{A} = \begin{pmatrix} \boldsymbol{B} & \boldsymbol{O} \end{pmatrix} \;\Rightarrow\; \boldsymbol{A}^{\#} = \begin{pmatrix} \boldsymbol{B}^{\#} \\ \boldsymbol{O}^T \end{pmatrix}, \qquad \boldsymbol{A} = \begin{pmatrix} \boldsymbol{B} \\ \boldsymbol{O} \end{pmatrix} \;\Rightarrow\; \boldsymbol{A}^{\#} = \begin{pmatrix} \boldsymbol{B}^{\#} & \boldsymbol{O}^T \end{pmatrix}, \tag{5}$$

$$\boldsymbol{u} \in \mathbb{R}^n \;\Rightarrow\; \boldsymbol{u}^{\#} = \left(\boldsymbol{u}^T\boldsymbol{u}\right)^{-1}\boldsymbol{u}^T = \frac{\boldsymbol{u}^T}{\|\boldsymbol{u}\|^2}, \qquad \left(\boldsymbol{u}^T\right)^{\#} = \frac{\boldsymbol{u}}{\|\boldsymbol{u}\|^2}. \tag{6}$$

The DK of the manipulator is $\frac{1}{2}\begin{pmatrix} C_1+C_{12}+C_{123}+C_{1234} \\ S_1+S_{12}+S_{123}+S_{1234} \end{pmatrix}$ so the Jacobian is

$$J = \frac{1}{2}\begin{pmatrix} -S_1-S_{12}-S_{123}-S_{1234} & -S_{12}-S_{123}-S_{1234} & -S_{123}-S_{1234} & -S_{1234} \\ C_1+C_{12}+C_{123}+C_{1234} & C_{12}+C_{123}+C_{1234} & C_{123}+C_{1234} & C_{1234} \end{pmatrix}$$ in the conf $q=0$:

$$J = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 2 & 3/2 & 1 & 1/2 \end{pmatrix}$$ to execute $v = (0 \quad 10)^T$ we need $J^{\#}$

We reorder the rows $\qquad B = (2 \quad 3/2 \quad 1 \quad 1/2)$

$$J = \begin{pmatrix} 2 & 3/2 & 1 & 1/2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow J^{\#} = \begin{pmatrix} B^{\#} & \begin{matrix} 0 \\ 0 \\ 0 \\ 0 \end{matrix} \end{pmatrix} = \frac{1}{15}\begin{pmatrix} 4 & 0 \\ 3 & 0 \\ 2 & 0 \\ 1 & 0 \end{pmatrix}$$

$$\Rightarrow J^{\#}v = \frac{1}{15}\begin{pmatrix} 4 & 0 \\ 3 & 0 \\ 2 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 10 \\ 0 \end{pmatrix} = \frac{1}{15}\begin{pmatrix} 40 & 30 & 20 & 10 \end{pmatrix}^T = \begin{pmatrix} 2.666 & 2 & 1.333 & 0.666 \end{pmatrix}^T \Rightarrow \dot{q}_3 \text{ exceeds } V_3 = 1$$

$\Rightarrow$ i scale for a factor $1/1.333 \Rightarrow \dot{q} = (2 \quad 1.5 \quad 1 \quad 0.5)^T$

Exercise 4

Consider a 3R planar robot with links of equal length $\ell = 1$ [m]. The primary task for this robot is to execute an instantaneous Cartesian velocity $\boldsymbol{v} \in \mathbb{R}^2$ with its end-effector. Denote the associated task Jacobian as $\boldsymbol{J}(\boldsymbol{q})$.

- When the robot is in the configuration $\boldsymbol{q}_0 = \begin{pmatrix} \pi/2 & \pi/3 & -2\pi/3 \end{pmatrix}^T$, use the Reduced Gradient (RG) method to determine the joint velocity $\dot{\boldsymbol{q}} \in \mathbb{R}^3$ that realizes a desired Cartesian velocity $\boldsymbol{v} = \begin{pmatrix} 1 & -\sqrt{3} \end{pmatrix}^T$ [m/s] while maximizing the objective function

$$H(\boldsymbol{q}) = \sin^2 q_2 + \sin^2 q_3. \tag{7}$$

- As an auxiliary task, the robot should move so as to always keep the position $\boldsymbol{p}_2 = (x_2, y_2)$ of the endpoint of its second link on the circle defined by

$$x_2^2 + (y_2 - 1.5)^2 = 0.75. \tag{8}$$

Determine the Jacobian $\boldsymbol{J}_a(\boldsymbol{q})$ associated to this auxiliary task. When the robot is in the configuration $\boldsymbol{q}_0$ defined above, is the robot in an algorithmic singularity? Can the two requested primary and auxiliary tasks be executed at the same time?

The Jacobian is

$$J = \begin{pmatrix} -S_{123}-S_{12}-S_1 & -S_{123}-S_{12} & -S_{123} \\ C_{123}+C_{12}+C_1 & C_{123}+C_{12} & C_{123} \end{pmatrix} \quad \text{In } q_0 \Rightarrow J(q_0) = \begin{pmatrix} -2 & -1 & -1/2 \\ 0 & 0 & \sqrt{3}/2 \end{pmatrix}$$

the gradient of H is: $\left(\frac{\partial H}{\partial q}\right)^T = \nabla H = \begin{pmatrix} 0 & 2C_2S_2 & 2C_3S_3 \end{pmatrix}^T$

We let $q_a = q_2, q_3$ and $q_b = q_1$ so:

$$J_a = \begin{pmatrix} -1 & -1/2 \\ 0 & \sqrt{3}/2 \end{pmatrix} \qquad J_b = \begin{pmatrix} -2 \\ 0 \end{pmatrix}$$

$$\nabla H'_{q_b} = \left(-(J_a^{-1}J_b)^T \quad I\right)\nabla H = \begin{pmatrix} -2 & 0 & 1 \end{pmatrix}\begin{pmatrix} 2C_2S_2 \\ 2C_3S_3 \\ 0 \end{pmatrix} = -4C_2S_2 = \dot{q}_b = -\sqrt{3} = \dot{q}_1$$

$$\dot{q}_a = J_a^{-1}(\dot{\xi} - J_b\dot{q}_b) = \begin{pmatrix} -1 & -0.577 \\ 0 & 1.1547 \end{pmatrix}\left(\begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix} - \begin{pmatrix} 2\sqrt{3} \\ 0 \end{pmatrix}\right) \Rightarrow \begin{cases} \dot{q}_2 = 3.4641 \\ \dot{q}_3 = -2 \end{cases}$$

$(\dot{q}_2 \; \dot{q}_3)$

Exercise 5

Consider a 2R planar robot with the nominal kinematics expressed by the DH parameters in Tab. 1.

i	α_i	d_i	a_i	θ_i
1	0	0	L_1	q_1
2	0	0	L_2	q_2

Table 1: Nominal kinematic parameters of a 2R planar robot.

In order to improve the robot accuracy, we would like to perform a large number of calibration experiments in which the robot end-effector position $\boldsymbol{p} \in \mathbb{R}^2$ (in the plane of motion) is measured by an external laser system. To recover the residual errors of the nominal direct kinematics with respect to the external measurements, it is assumed that we require only the fine tuning of the parameters $\boldsymbol{a} \in \mathbb{R}^2$ and of the encoder measurements $\boldsymbol{\theta} \in \mathbb{R}^2$, whose adjustments will be performed simultaneously. What will be the expression of the 2×4 regressor matrix $\boldsymbol{\Phi}$ for a single calibration experiment?

the DK is $\mathcal{K} = \begin{cases} a_1 c_1 + a_2 c_{12} \\ a_1 s_1 + a_2 s_{12} \end{cases}$ so $\Rightarrow \dfrac{\partial \mathcal{K}}{\partial a} = \begin{pmatrix} c_1 & c_{12} \\ s_1 & s_{12} \end{pmatrix}$ and $\dfrac{\partial \mathcal{K}}{\partial q} = J = \begin{pmatrix} -a_1 s_1 - a_2 s_{12} & -a_2 s_{12} \\ a_1 c_1 + a_2 c_{12} & a_2 c_{12} \end{pmatrix}$

$$\Rightarrow \Phi = \left(\frac{\partial \mathcal{K}}{\partial a} \quad J \right) = \begin{pmatrix} c_1 & c_{12} & -a_1 s_1 - a_2 s_{12} & -a_2 s_{12} \\ s_1 & s_{12} & a_1 c_1 + a_2 c_{12} & a_2 c_{12} \end{pmatrix} \qquad \begin{aligned} \theta &= (q_1 \;\; q_2)^T \\ a &= (L_1 \;\; L_2)^T \end{aligned}$$

For a single cal. we solve $\Delta \mathcal{Y} = \Phi \Delta \phi$

error in measurement

error in θ, a